Gayatri Vidya Parishad College of Engineering Autonomous

Civil Engineering | Course Code: CE201

Surveying & Geomatics - Laboratory Manual & Practical Guide

Syllabus Focus: CE201 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Linear Measurements & Compass Surveying

" Syllabus Scope: Chain and tape surveying, Magnetic declination, Whole circle bearing vs reduced bearing

Unit 2: Levelling & Contouring

" Syllabus Scope: Differential levelling, Collimation and Rise-and-Fall computation methods, Reciprocal l

CE201: Surveying & Geomatics - Advanced Topics & Assessment

Academic Track: Civil Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Theodolite Traversing & Computation

" Vernier transit theodolite, Measurement of horizontal and vertical angles, Gale traverse table, Bowditch and

Unit 4: Curve Setting Horizontal & Vertical

" Circular and transition curves, Elements of simple curves, Setting out curves by offsets from chord produced

Unit 5: Modern Geomatics & Total Station

" Electronic Distance Measurement EDM, Total Station operations, Global Positioning System GPS/GNSS surveying,

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Differential Levelling and Fly Levelling Exercise across Building Benchmarks

Lab 2: Measurement of Horizontal Angles by Repetition and Reiteration Methods using Vernier Theodolite

Lab 3: Setting Out a Simple Circular Curve on the Field using Rankine Deflection Angle Method

Lab 4: Topographical Surveying and Area Computation using Electronic Total Station

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* B.C. Punmia, Ashok Kumar Jain - Surveying Vol. I & II Laxmi Publications

* K.R. Arora - Surveying Vol. 1 & 2 Standard Book House

Authorized NoteVault Educational Document " Verified for Registered University Students